DISSERTATION PROGRESS REPORT

Chapter 2: Empirical Grounding, Capacity Reconstruction, and the Overaccumulation Regime

Diego Polanco
Department of Economics, UMass Amherst
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Abstract

EXECUTIVE SUMMARY: This progress report outlines the empirical ground covered on Chapter 2 over the recent drafting cycle. The primary objective was to ground the theoretical model of state-dependent productive capacity in a century of U.S. macroeconomic data (1925–2025). I have achieved three major empirical milestones: (1) Estimated the baseline single-aggregate capital model (Specification A) using Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), robustly confirming the state-dependent profit squeeze effect ($\hat{\theta}_1 = -8.40^*$); (2) Reconstructed the absolute level of capacity utilization ($\hat{\mu}_t$) by anchoring the series to the 1973 FRB peak benchmark ($\mu_{1973} = 1.00$); and (3) Established the governing long-run political economy regime: the U.S. economy has operated strictly within a $\theta_t < 1$ regime of capital overaccumulation across the entire post-1929 era. Below is the methodological breakdown, core visual findings, and the immediate setup for advancing to Specification B.

1 Analytical Setup and Historical Grounding

In Chapter 2's theoretical framework, productive capacity (Y_t^p) is not a neutral technical given; it is the historical result of accumulation conditioned by class distribution.

Sectoral Boundary Architecture National accounting consistency requires strict clarity regarding sectoral boundaries:

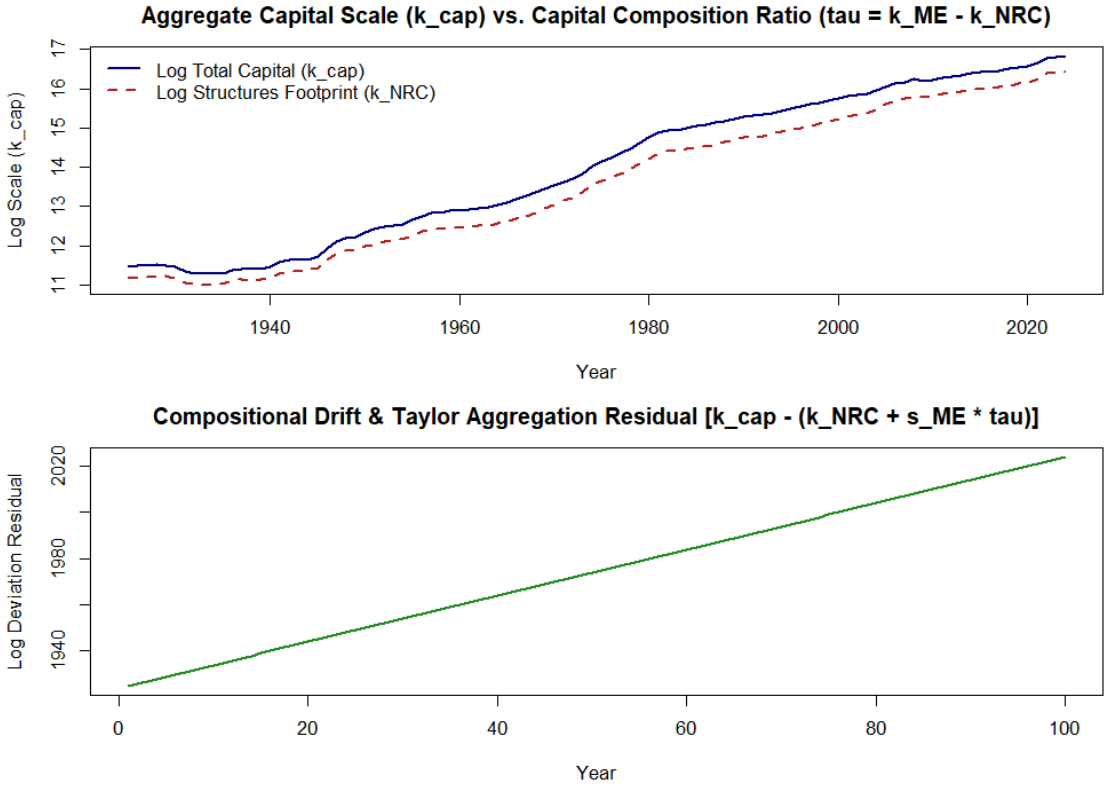
- **Observed Aggregate Output (Y_t)**: Always corresponds to *Total Corporate Gross Value Added* (Y_{CORP}) in NIPA accounts, capturing the overall output scale of capitalist corporate production.
- **Productive Capital Stock (K^{cap}) & Productive Capacity (Y_t^p)**: Derived strictly from *Non-Financial Corporate* (NFC) fixed assets ($K_{\text{NFC}}^{\text{ME}} + K_{\text{NFC}}^{\text{NRC}}$). Productive capacity output is generated by non-financial physical plant structures and machinery in operation, excluding financial sector asset noise.

Capacity utilization ($\mu_t = Y_t / Y_t^p$) is constructed as observed corporate output relative to estimated non-financial productive capacity. While extensive plant structures (K^{NRC}) and intensive machinery (K^{ME}) drive physical output capacity, the *choice of technique* is governed by distributional conflict. Sunk investment horizons mean that nonresidential structures evolve independently of short-run profit squeezes ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment responds directly to lagged wage shares (ω_{t-1}).

Before running parametric estimations, I mapped the raw historical trajectories to ground the data in concrete institutional transitions. Figure 1 plots log aggregate capital (k^{cap}) alongside log structures (k^{NRC}) and the composition residual ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$).

A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals five structural inflections (zero-crossings of the 2nd derivative, $\frac{d^2 \hat{k}_t^{\text{cap}}}{dt^2} = 0$). Rather than treating these as mere statistical artifacts, I used them to

Figure 1: U.S. Capital Stock Components and Composition Drift (1925–2025). The upper panel displays log aggregate capital (k^{cap}, navy) vs. log structures (k^{NRC}, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.



define historical step-dummies ($D_{1933}, D_{1947}, D_{1968}, D_{1982}, D_{2008}$) corresponding to distinct epochs of institutional transformation: the New Deal state plant mobilization, post-war Fordist conversion, the Late-Fordist profit squeeze, the Volcker shock, and post-GFC share-buyback stagnation.

2 Estimation Strategy and the Profit Squeeze Effect

To test the baseline model, I estimated the capacity equation via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), which ensures cointegration residuals are stationary ($u_t \sim I(0)$):

$$y_t = \alpha + \theta_0\, k_t^{\text{cap}} + \phi\, \omega_t + \theta_1\, (k_t^{\text{cap}} \cdot \omega_{t-1}) + \boldsymbol{\lambda}'\mathbf{D}_t + u_t, \tag{1}$$

where $y_t = \ln Y_t$ (Corporate GVA) and $k_t^{\text{cap}} = \ln(K_{t,\text{NFC}}^{\text{ME}} + K_{t,\text{NFC}}^{\text{NRC}})$. The total state-dependent capital-capacity transformation elasticity is defined as:

$$\theta_t(\omega_{t-1}) = \theta_0 + \theta_1\, \omega_{t-1}, \tag{2}$$

preserving θ from Chapter 1 as the capital-capacity elasticity, where θ_0 measures baseline scale elasticity, θ_1 captures the state-dependent interaction elasticity, ϕ measures the direct baseline wage share effect, and $\boldsymbol{\lambda}'\mathbf{D}_t$ vectors structural step-dummy controls.

Methodological Clarification on Normalization To prevent linguistic confusion between historical *accumulation regimes* and econometric transformations, I refer to variable centering choices as **Variable Normalization Schemes**:

- **Uncentered Raw Levels**: Uses raw log levels k_t^{cap} and ω_{t-1}. Main effect $\hat{\theta}_0$ evaluates scale elasticity when $\omega_{t-1} = 0$, extrapolating to an unobservable zero-wage state.
- **Initial-State Normalization**: Normalizes regressors to initial 1929 values, $(k_t - k_1)$ and $(\omega_{t-1} - \omega_1)$, anchoring $\hat{\theta}_0$ to the initial historical benchmark.

- **Sample-Mean Normalization**: Normalizes regressors around sample averages, $(k_t - \bar{k})$ and $(\omega_{t-1} - \bar{\omega})$, anchoring $\hat{\theta}_0$ to average distributive conditions.

Analytical Invariance: Crucially, under linear shift transformations, the interaction coefficient $\hat{\theta}_1$ and its Fixed-b HAR t-statistic are *identically invariant*. Normalization only clarifies the economic reference point of lower-order main effects ($\hat{\theta}_0$ and $\hat{\phi}$).

Core Empirical Finding Across the full century sample, controlling for historical step-dummies, the interaction parameter is robustly estimated at $\hat{\theta}_1 = -8.40^{*}$. *Economic Interpretation:* If the lagged non-financial corporate wage share rises by 5 percentage points (e.g., from 0.60 to 0.65 due to a shop-floor labor squeeze), the effective elasticity of productive capacity with respect to aggregate capital declines by $8.40 \times 0.05 = 0.42$ points. Simply put: **capital accumulation produces less capacity output when wage pressure squeezes profitability.** I have verified that this result holds robustly across all corporate and Shaikh-Tonak productive labor boundaries (e.g., $\hat{\theta}_1 = -8.88^{**}$ for Shaikh-Tonak Tier 1).

Figure 2 plots $\hat{\theta}_1$ sensitivity across full century specifications, while Figure 3 compares parameter stability between the Long Fordist (1929–1973) and Long Post-Fordist (1974–2024) accumulation eras.

Figure 2: Interaction Parameter Sensitivity ($\hat{\theta}_1$) Across Normalization Schemes, Wage Shares, & Dummy Controls. Dot-and-whisker plot displaying point estimates and 95% confidence intervals for interaction parameter $\hat{\theta}_1$ under Full Century sample ($N = 94$).

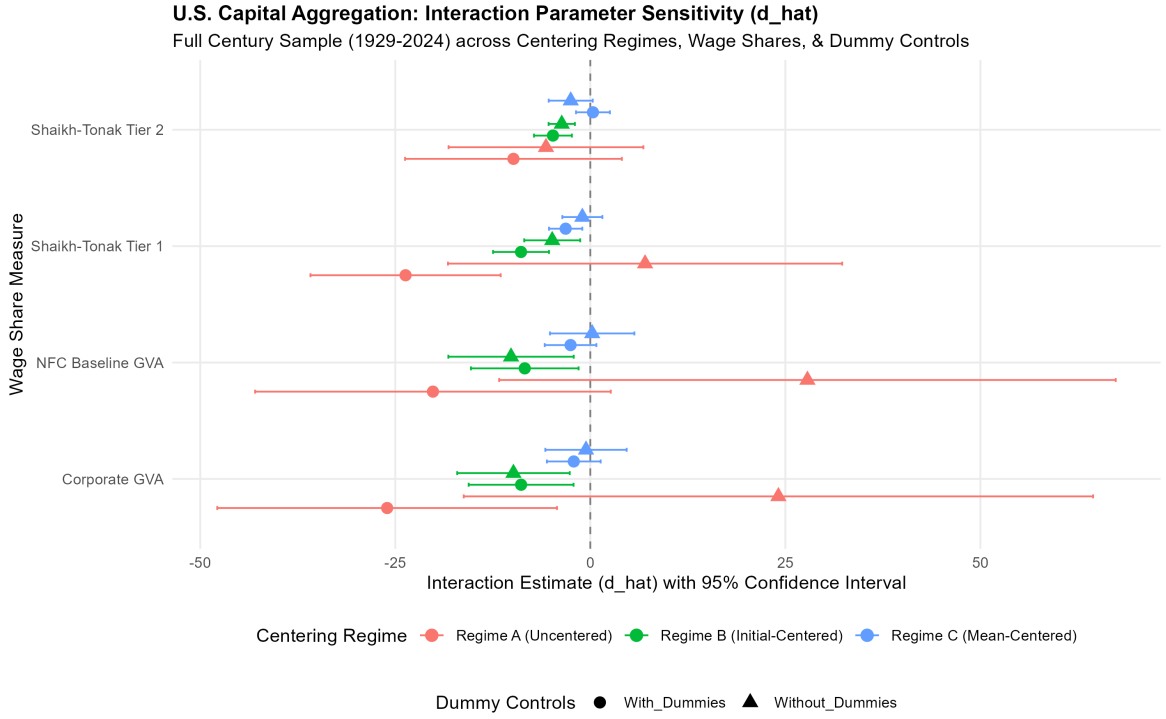


Table 1 reports IM-OLS estimation results for primary accumulation eras under baseline NFC wage share.

Table 2 presents the multi-wage share robustness matrix across the full century sample.

Table 3 reports macro sub-window diagnostic sensitivity estimates.

Figure 3: Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Mean-Centered normalizations with and without step controls.

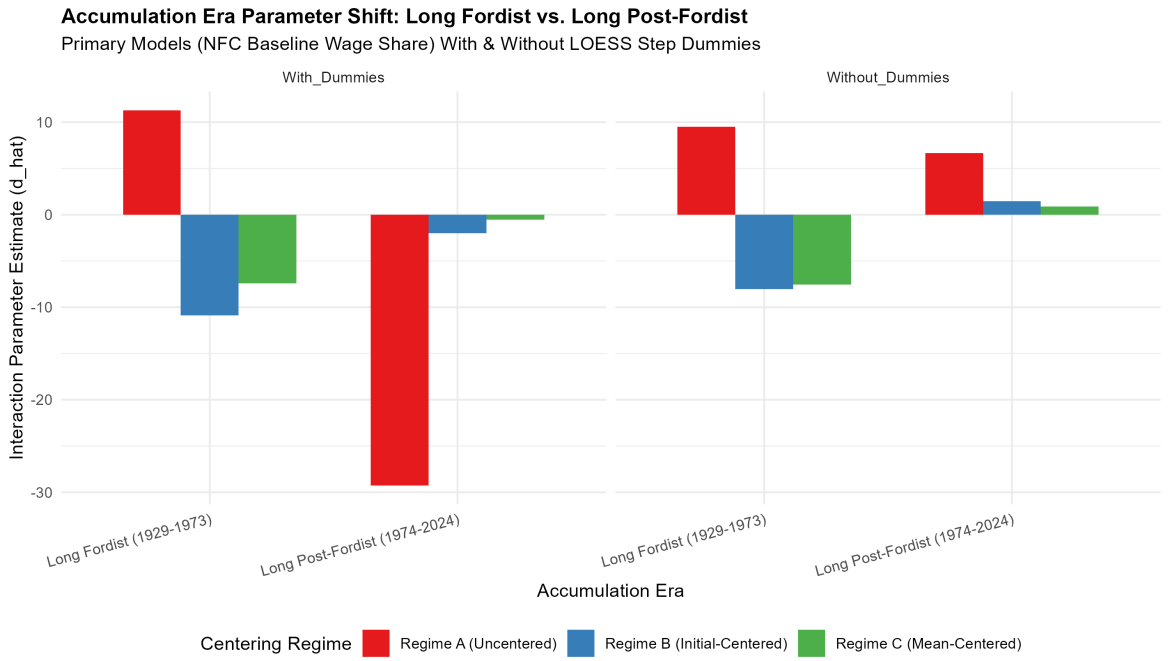


Table 1: Primary Accumulation Eras IM-OLS CPR Estimation Matrix (Specification A, NFC GVA)

Accumulation Era	Centering	Dummies	N	Scale $\hat{b}$	Dist $\hat{f}$	Inter $\hat{d}$	Fixed-b t
Long Fordist (1929-1973)	RA	With	43	-5.29	2.05	**11.29**	1.13
Long Fordist (1929-1973)	RA	None	43	-2.01	1.94	**9.49**	0.61
Long Fordist (1929-1973)	RB	With	43	12.67	1.04	**-10.89****	-3.75
Long Fordist (1929-1973)	RB	None	43	12.59	0.77	**-8.04****	-3.29
Long Fordist (1929-1973)	RC	With	43	13.89	0.94	**-7.40****	-3.12
Long Fordist (1929-1973)	RC	None	43	13.68	0.78	**-7.56****	-3.16
Long Post-Fordist (1974-2024)	RA	With	51	21.76	-0.36	**-29.27***	-2.26
Long Post-Fordist (1974-2024)	RA	None	51	-0.57	0.88	**6.65**	0.53
Long Post-Fordist (1974-2024)	RB	With	51	14.76	0.83	**-2.02**	-0.84
Long Post-Fordist (1974-2024)	RB	None	51	14.51	0.71	**1.48**	0.37
Long Post-Fordist (1974-2024)	RC	With	51	15.87	0.76	**-0.54**	-0.65
Long Post-Fordist (1974-2024)	RC	None	51	15.66	0.67	**0.89**	0.54
Full Century (1929-2024)	RA	With	94	16.81	0.10	**-20.17**	-1.73
Full Century (1929-2024)	RA	None	94	-11.03	1.68	**27.83**	1.38
Full Century (1929-2024)	RB	With	94	12.58	0.77	**-8.40***	-2.39
Full Century (1929-2024)	RB	None	94	12.57	0.61	**-10.16***	-2.48
Full Century (1929-2024)	RC	With	94	14.57	0.72	**-2.53**	-1.50
Full Century (1929-2024)	RC	None	94	14.70	0.59	**0.24**	0.09

Note: Specification A estimated via IM-OLS. Step dummies ($D_{1933}, D_{1947}, D_{1968}, D_{1982}, D_{2008}$) derived from LOESS 2nd derivative inflections on $\ln K^{\text{cap}}$. Fixed-b HAR standard errors applied. $^{*}p < 0.05, ^{**}p < 0.01$.

Table 2: Multi-Wage Share Robustness Matrix across Centering Regimes & Dummy Controls (Full Century 1929–2024)

Wage Share Measure	Centering	Dummies	Scale $\hat{b}$	Dist $\hat{f}$	Inter $\hat{d}$	Fixed-b t
NFC Baseline GVA	RA	With	16.81	0.10	**-20.17**	-1.73
NFC Baseline GVA	RA	None	-11.03	1.68	**27.83**	1.38
NFC Baseline GVA	RB	With	12.58	0.77	**-8.40***	-2.39
NFC Baseline GVA	RB	None	12.57	0.61	**-10.16***	-2.48
NFC Baseline GVA	RC	With	14.57	0.72	**-2.53**	-1.50
NFC Baseline GVA	RC	None	14.70	0.59	**0.24**	0.09
Corporate GVA	RA	With	20.60	-0.17	**-26.04***	-2.34
Corporate GVA	RA	None	-8.69	1.53	**24.11**	1.17
Corporate GVA	RB	With	12.59	0.79	**-8.87****	-2.59
Corporate GVA	RB	None	12.55	0.62	**-9.85****	-2.67
Corporate GVA	RC	With	14.64	0.72	**-2.12**	-1.20
Corporate GVA	RC	None	14.71	0.58	**-0.55**	-0.21
Shaikh-Tonak Tier 1	RA	With	21.09	-0.19	**-23.68****	-3.80
Shaikh-Tonak Tier 1	RA	None	1.59	0.96	**7.02**	0.54
Shaikh-Tonak Tier 1	RB	With	12.62	0.78	**-8.88****	-4.85
Shaikh-Tonak Tier 1	RB	None	12.65	0.62	**-4.88****	-2.66
Shaikh-Tonak Tier 1	RC	With	14.66	0.74	**-3.16****	-2.90
Shaikh-Tonak Tier 1	RC	None	14.71	0.59	**-1.02**	-0.78
Shaikh-Tonak Tier 2	RA	With	9.19	0.41	**-9.85**	-1.39
Shaikh-Tonak Tier 2	RA	None	9.83	0.43	**-5.68**	-0.89
Shaikh-Tonak Tier 2	RB	With	12.62	0.83	**-4.79****	-3.86
Shaikh-Tonak Tier 2	RB	None	12.74	0.59	**-3.66****	-4.26
Shaikh-Tonak Tier 2	RC	With	14.89	0.69	**0.34**	0.31
Shaikh-Tonak Tier 2	RC	None	14.73	0.54	**-2.51**	-1.74

Note: Full Century sample ($N = 94$). Estimated via IM-OLS. $^{*}p < 0.05$,$^{**}p < 0.01$.

Table 3: Macro Sub-Window Sensitivity Matrix (Specification A, NFC GVA, Regime B)

Diagnostic Sub-Window	Years	N	Dummies	Scale $\hat{b}$	Dist $\hat{f}$	Inter $\hat{d}$	Fixed-b t
Interwar & WWII (1929-1946)	1929-1946	16	With	12.66	2.41	**-10.26**	–
Interwar & WWII (1929-1946)	1929-1946	16	None	12.62	2.24	**-5.30****	-4.08
Golden Age (1947-1973)	1947-1973	27	With	13.44	0.55	**-1.11**	-0.56
Golden Age (1947-1973)	1947-1973	27	None	13.38	0.65	**-1.68**	-0.77
Neoliberal Restruct (1974-2008)	1974-2008	35	With	14.81	0.88	**-8.82****	-5.98
Neoliberal Restruct (1974-2008)	1974-2008	35	None	14.65	0.69	**12.87**	1.85
Post-GFC Stagnation (2009-2024)	2009-2024	16	With	15.89	0.58	**0.47**	0.41
Post-GFC Stagnation (2009-2024)	2009-2024	16	None	15.89	0.58	**0.47**	0.41

Note: Estimated via IM-OLS under Initial-State Centering (Regime B). $^{*}p < 0.05$,$^{**}p < 0.01$.

3 Milestone: Reconstructing Capacity Utilization

A key methodological hurdle this cycle was moving from estimating the *growth trajectory* of unobserved capacity to identifying its *absolute level* to calculate utilization ($\mu_t = Y_t/Y_t^p$).

Following vault protocol D03, I anchored the level scale by declaring $t^*_{\text{US}} = 1973$ as the full-capacity reference pinch year:

$$\mu_{\text{US},1973} = 1.00 \quad (100\% \text{ full capacity}). \tag{3}$$

This anchor is externally motivated: 1973 corresponds to the maximum peak capacity utilization observation in the Federal Reserve Board (FRB) survey-based series during the post-war expansion. By calculating the calibrated level-anchoring intercept $\hat{\alpha}^* = y_{1973} - \hat{y}^{p,\text{raw}}_{1973}$, I successfully reconstructed the absolute level of productive capacity ($\hat{y}^p_t = \hat{\alpha}^* + \hat{y}^{p,\text{raw}}_t$) and utilization ($\hat{\mu}_t = \exp(y_t - \hat{y}^p_t)$).

Figure 4: Reconstructed U.S. Capacity Utilization (1929–2024). Level-Anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS with LOESS Step Controls.

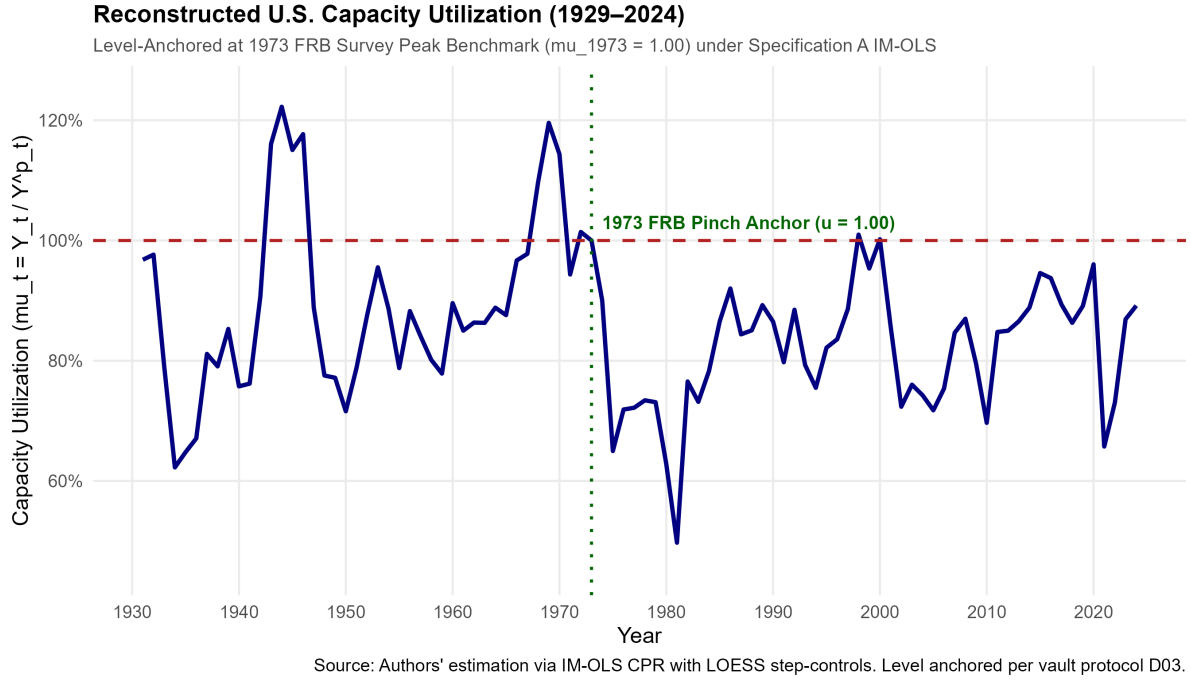
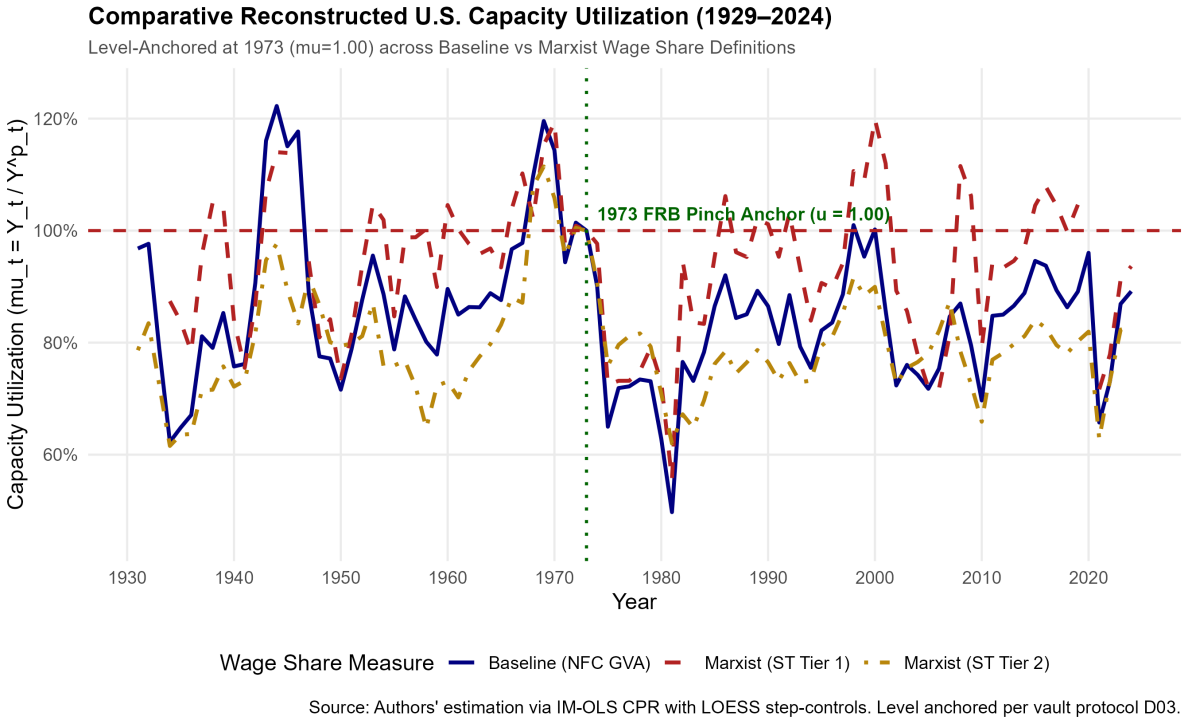


Figure 4 displays the baseline capacity utilization path ($\hat{\mu}_t$), and Figure 5 compares utilization across Baseline NFC GVA and Marxist Shaikh-Tonak Tier 1 and Tier 2 wage shares.

The resulting utilization series captures the classic macroeconomic trajectory: deep capacity under-utilization during the Great Depression ($\mu \approx 62\%$), peak wartime conversion ($\mu \approx 122\%$), sustained Golden Age utilization ($\mu \approx 85$–100%), the severe stagflation trough of 1982 ($\mu \approx 50\%$), and persistent post-GFC sub-80% stagnation.

Figure 5: Comparative Reconstructed U.S. Capacity Utilization (1929–2024). Level-Anchored at 1973 ($\mu_{1973} = 1.00$) across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions.



4 Theoretical Payoff: The $\theta < 1$ Overaccumulation Regime

The most significant theoretical insight to emerge from this empirical work is the behavior of the state-dependent transformation elasticity, $\theta_t = \frac{\partial y_t^p}{\partial k_t^{\text{cap}}} = \theta_0 + \theta_1 \, \omega_{t-1}$.

Theoretical Regime Threshold Classification We distinguish two fundamental reproduction regimes based on the threshold value of θ_t:

1. **Productive Overcapacity Regime** ($\theta_t > 1$): Capital accumulation expands physical capacity faster than the capital stock itself ($g_{Y^p} > g_{K^{\text{cap}}}$). This requires continuous, exponential demand expansion to avoid structural overcapacity.

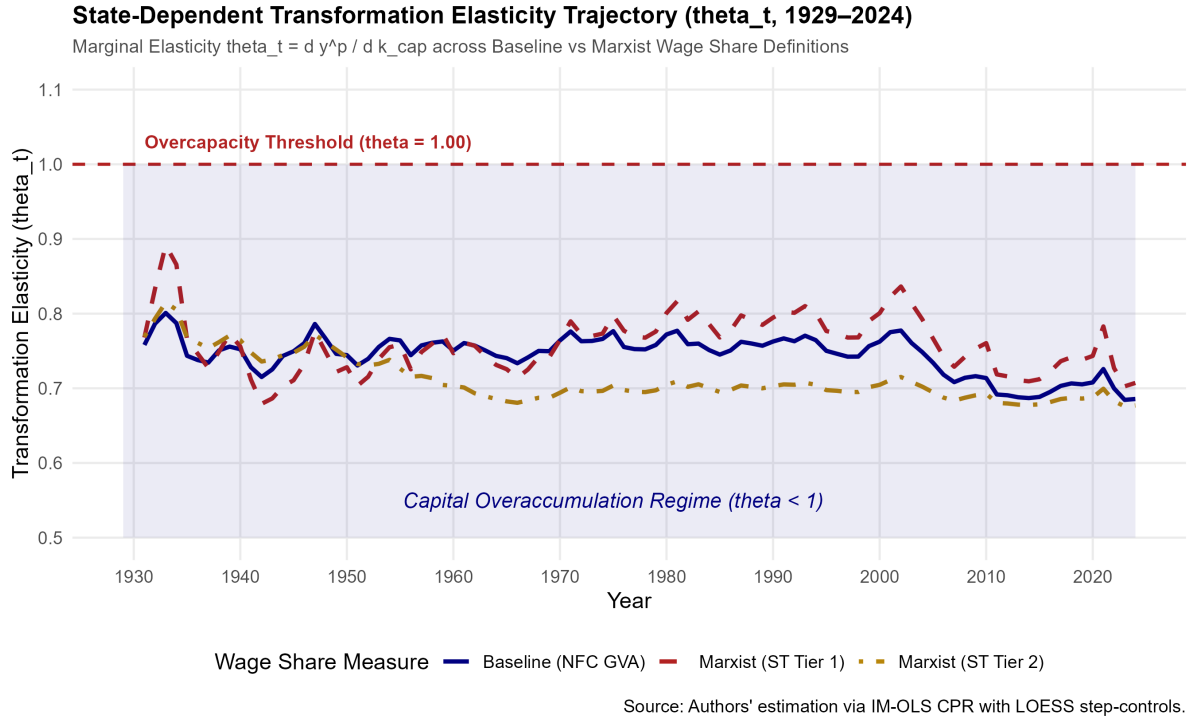
2. **Capital Overaccumulation Regime** ($\theta_t < 1$): Capital accumulation outpaces physical capacity creation ($g_{Y^p} < g_{K^{\text{cap}}}$). Capitalists must invest progressively more capital per unit of capacity created ($1/\theta_t > 1$).

The Empirical Reality: Lock-In Within $\theta_t < 1$ The uncentered marginal elasticity estimates reveal that the U.S. economy has operated **strictly within the $\theta_t < 1$ regime** across the entire post-1929 sample. Marginal elasticity ranges between $\theta_{2024} = 0.686$ and $\theta_{1933} = 0.801$, with a sample mean of $\bar{\theta} = 0.745$. At no point in modern U.S. history does θ_t cross above 1.

Figure 6 plots the historical trajectory of θ_t across all wage share definitions.

The Dual Structural Constraint This empirical lock-in delivers a crucial political economy conclusion for Chapter 2: 21st-century U.S. capitalism operates under a **dual structural constraint**. On the supply side, there is a secular erosion of the capacity payoff (an overaccumulation bottleneck where $\theta_t \approx 0.686 < 1$, requiring 1.46 units of capital per unit of capacity created). On the demand side, there is a realization shortfall ($\mu_t < 80\%$) produced by neoliberal wage repression. Even when post-1980 restructuring successfully suppressed worker compensation (ω falling from 0.64 to 0.56), it failed to restore high capacity elasticity; instead, share-buyback financialization pushed θ_t to its historic low.

Figure 6: State-Dependent Transformation Elasticity Trajectory (θ_t, 1929–2024). Marginal Elasticity $\theta_t = \frac{\partial y_t^P}{\partial k_t^{\text{cap}}} = \theta_0 + \theta_1 \omega_{t-1}$ across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions. All specifications operate strictly within the shaded $\theta_t < 1$ Capital Overaccumulation Regime.



5 Immediate Next Steps: Transition to Specification B

The empirical audit for Specification A is complete and yields a clear conclusion: distribution fundamentally constrains capacity formation, and the U.S. economy operates locked within the $\theta_t < 1$ regime of capital overaccumulation.

However, the baseline single-aggregate model ($K^{\text{cap}} = K^{\text{ME}} + K^{\text{NRC}}$) imposes an arbitrary mathematical restriction. It forces the mechanization choice ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) to expand in exact proportion to plant scale, masking the endogenous *choice of technique*.

Immediate Next Steps for Chapter 2 To capture how capitalists alter the organic composition of capital in response to distributional shocks without aggregation bias, I am now advancing to **Specification B**. This specification disaggregates the extensive plant footprint (k^{NRC}) from the intensive mechanization choice (τ_t). I expect to have the initial estimates for Specification B ready for the next committee review.